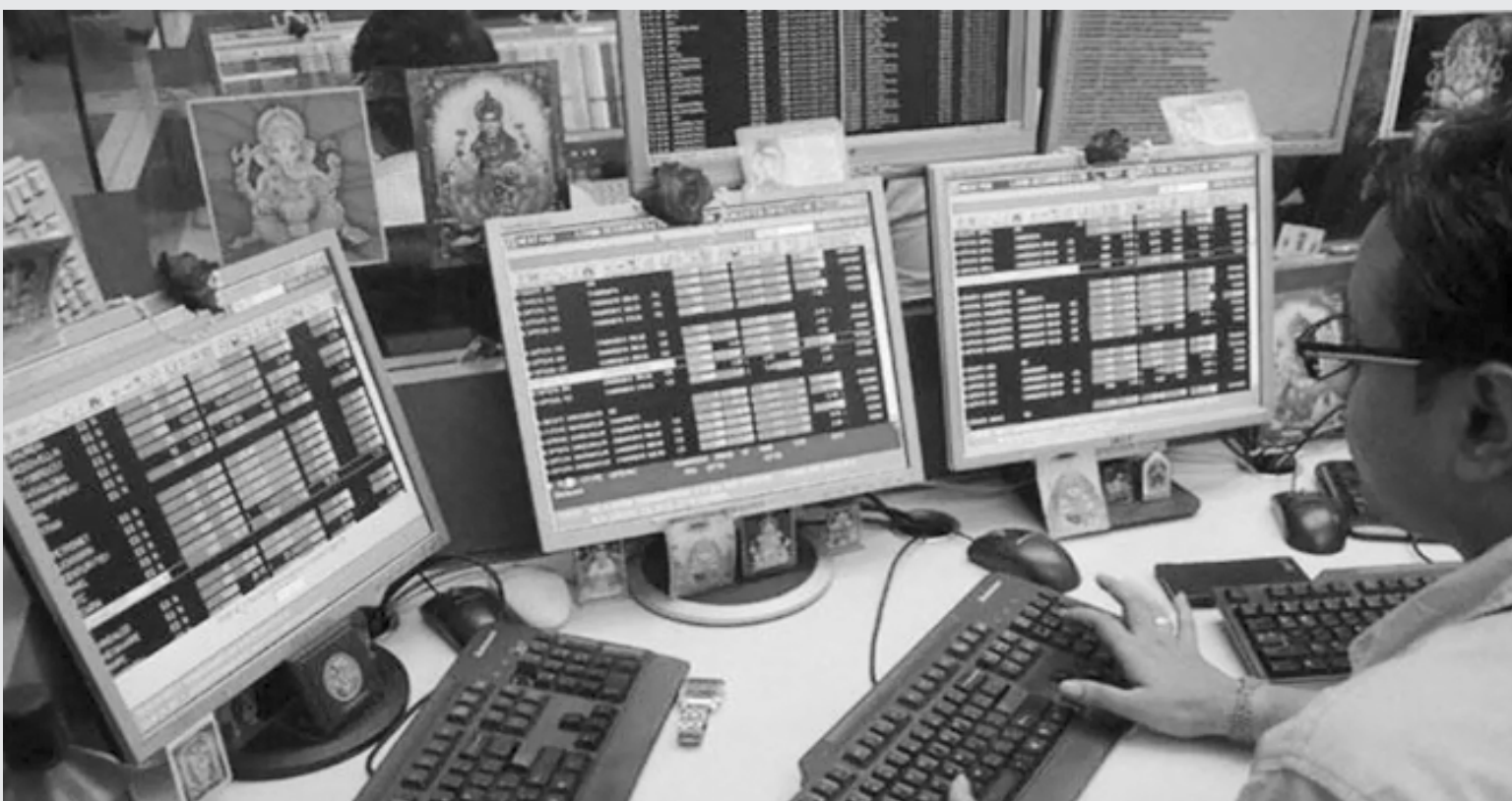


12

Statistics



The financial market is completely reliant on statistical analysis. It is used to compute all the stock prices. It also assists the investor in deciding whether to invest in a certain stock or not. It also aids corporations in managing their finances in order to conduct long-term operations.



What's Inside

Topic(s)

- ✓ Topic-wise Notes
- ✓ Related NCERT Qs
- ✓ Caution Points
- ✓ Case-based Examples

Questions

- ✓ New Pattern Qs
- ✓ All Objective + Subjective Qs
- ✓ Diksha + Exemplar Qs
- ✓ Detailed Explanation

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| TOPIC 1 |

GRAPHICAL REPRESENTATION OF DATA

Learning Objectives

- Students will be learning the different geometrical methods of representing the data.

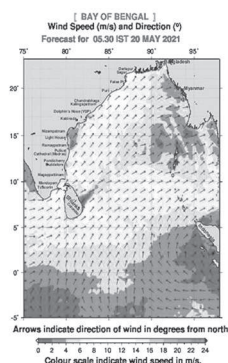
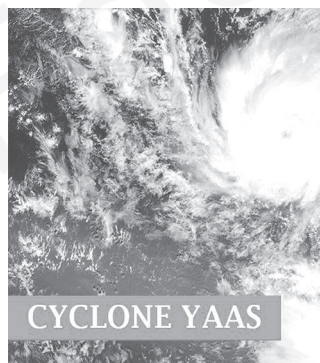
Learning Outcomes

- Students will be able to represent the data on bar graph, histogram and frequency polygon.

Real Life Application

Amid the second wave of coronavirus, India witnessed two cyclones in May 2021: 'Tauktae' and 'Yaas' in the Gujarat and West Bengal, respectively. We hear about several disasters, some of which may occur in the future. The Indian Meteorological Department alerts the citizens before it happens and also announces which places they can affect.

So, have you ever wondered how meteorologists inform us about upcoming disasters? All these alerts and announcements they do with the help of graphical representation of statistics.

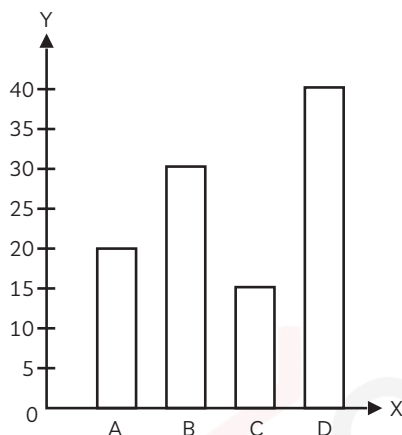


Meteorologists use many different tools for measuring temperature, air pressure, and wind speed in all the layers of the troposphere. They collect all the data and plugg it into the graphs. So, graphical representation of data is always applied in prediction of natural disasters.

Using graph, the researchers and analysts configure ways for optimum utilisation of the available resources and formulate strategies to overcome the possible side-effects of every particular disaster individually. Let's now learn graphical representation in detail.

Bar Graph

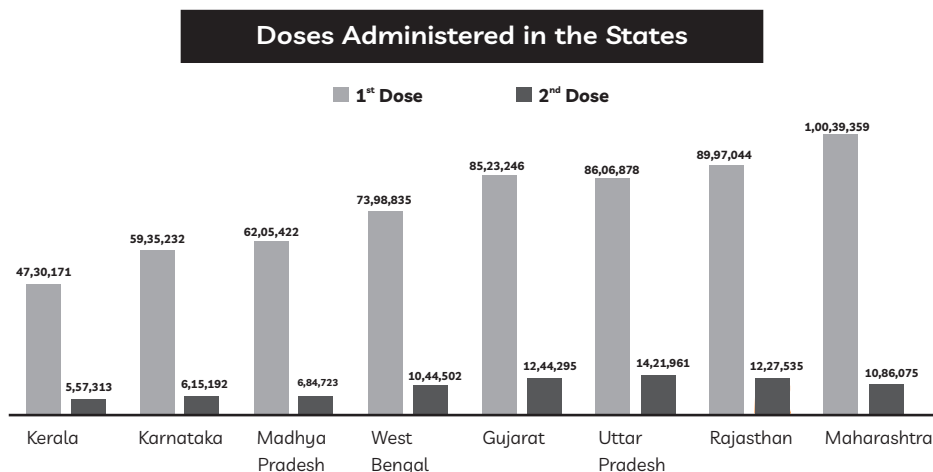
A bar graph is a pictorial representation of data in which usually bars of uniform width are drawn with equal spacing between them on one axis (say, the x-axis), depicting the variable. The values of the variable are shown on the other axis (say, the y-axis) and the heights of the bars depend on the values of the variable.



In the pandemic of COVID-19, to collect the data of the vaccination program state-wise, and to determine the number of people who got 1st dose and who got both doses. Graphs played a very crucial role to understand the vaccination program at various stages.

Graphical representation of data is a visual representation and it is more often used and effective than representing the data in a tabular form. Since, graphs are powerful abstractions, they can be essential in modeling data.

Nowadays, the spread of disease is widespread worldwide. Analysing such graphs has become an essential component in understanding and controlling the spread of diseases.



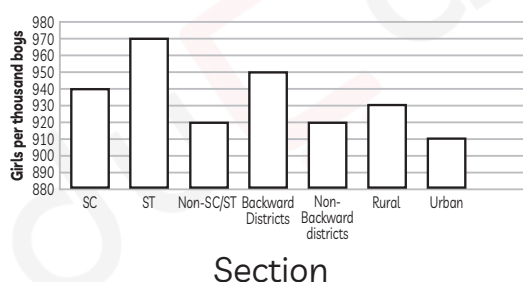
Example 1. The following data on the number of girls (to the nearest ten) per thousand boys in different sections of Indian society is given below.

Section	Number of girls per thousand boys
Scheduled Caste (SC)	940
Scheduled Tribe (ST)	970
Non-SC/ST	920
Backward districts	950
Non-Backward districts	920
Rural	930
Urban	910

(A) Represent the above information by a bar graph.

(B) In the classroom, discuss what conclusion can be arrived at from the drawn graph. [NCERT]

Ans. (A) The bar graph is as follows:



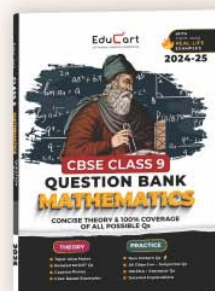
(B) From above, we can conclude:

- (1) The maximum number of girls per thousand boys (i.e., 970) is for ST.
- (2) A minimum number of girls per thousand boys (i.e., 910) is for urban areas.



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OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. In a continuous frequency distribution, class mark of a class is 85 and lower limit is 83, then its upper limit is:

- (a)80 (b)89
(c)86 (d)87

[Diksha]

Ans.(d)87

Explanation:

$$\text{Class mark} = \frac{\text{upper limit} + \text{lower limit}}{2}$$

$$85 = \frac{\text{upper limit} + 83}{2}$$

$$\text{Upper limit} = 85 \times 2 - 83$$

$$\text{Upper limit} = 170 - 83 = 87$$

Assertion and Reason (A–R)

Direction for the given question: A statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
(c) Assertion (A) is true but reason (R) is false.
(d) Assertion (A) is false but reason (R) is true.

2. Assertion (A):The range of first 9 multiple of 9 is 27.

Reason (R):Range = Highest value – Lowest value.

Ans. (d) Assertion (A) is false but reason (R) is true.

Explanation: First 9 multiples of 9 are 9, 18, 27, 36, 45, 54, 63, 72, 81

We have,

Range = Highest value – Lowest value

$$\therefore \text{Range} = 81 - 9 = 72$$

Hence, assertion (A) is false, but reason (R) is true.



CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

3. A foreign direct investment is an investment in the form of controlling ownership in a business in one country by an entity based in another country. India is a major monetary source for economic development in India. Foreign companies invest directly in fast growing private Indian businesses to take benefits of cheaper wages and changing business environment of India. The bar graph diagram shows the trends of foreign direct investment (FDI) into a public sector undertakings in India from all over the world in 2016–2022. (In crores)



- (A) The sum of FDI in 2016 and 2017, is:
(a) 18.58 cr (b) 15.85 cr
(c) 15.22 cr (d) 15.75 cr
- (B) The year which exhibited the highest FDI in the PSU of India over the period, is:
(a) 2017 (b) 2018
(c) 2020 (d) 2021
- (C) The absolute difference in FDI in the PSU of India between 2020 and 2021, is:
(a) 7.29 cr (b) 7.13 cr
(c) 7.16 cr (d) 7.22 cr

(D) The year which exhibited the lowest FDI in the PSU of India over the period is:

(a) 2017 (b) 2020

(c) 2016 (d) 2015

Ans. (A)(b) 15.85 cr

Explanation: From the given bar graph,

The sum of FDI in 2016 and 2017

$$= 5.7 + 10.15$$

$$= 15.85 \text{ cr}$$

(B)(d) 2021

Explanation: From the graph, it is clear that in 2021, the PSU got the highest FDI.

(C)(b) 7.13 cr

Explanation: FDI in 2020 = 24.23 cr

FDI in 2021 = 31.36 cr

Absolute difference in the given period = $(31.36 - 24.23)$ cr

$$= 7.13 \text{ cr}$$

(D)(c) 2016

Explanation: from the given graph, in 2016, the PSU got lowest FDI that is 5.7 cr.

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

4. The points scored by a basketball team in a series of matches are as follows:

17, 7, 10, 25, 5, 10, 18, 10 and 24.

Find the range of the points.

Ans. Here, maximum points = 25 and minimum points = 5

Range = Maximum value – Minimum value

$$= 25 - 5 = 20$$

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

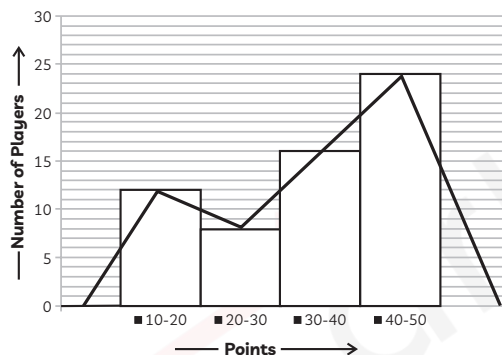


5. The table below shows the points scored by 60 players.

Points	Number of players
10-20	12
20-30	8
30-40	16
40-50	24

Make the frequency polygon.

Ans.



SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

6. Draw a histogram and a frequency polygon for the following frequency distribution.

Length (in mm)	Frequency
1 – 10	8
11 – 20	3
21 – 30	6
31 – 40	12
41 – 50	2
51 – 60	7

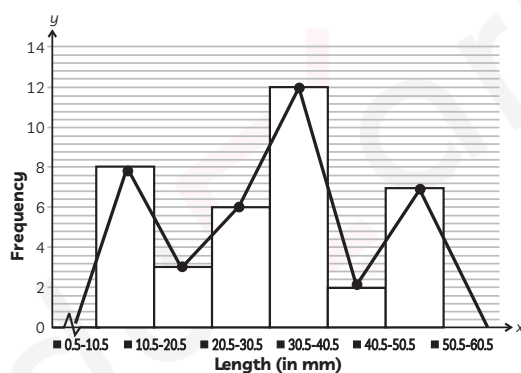
Ans. As data is not continuous, so to make a histogram we need continuous data.

$$\text{Gap} = 11 - 10 = 1$$

$$\text{So, Half of gap} = \frac{1}{2} = 0.5$$

∴ We add 0.5 to each upper class limit and subtract 0.5 from each lower class limit to make the class intervals continuous.

Length (in mm)	Frequency
0.5 – 10.5	8
10.5 – 20.5	3
20.5 – 30.5	6
30.5 – 40.5	12
40.5 – 50.5	2
50.5 – 60.5	7



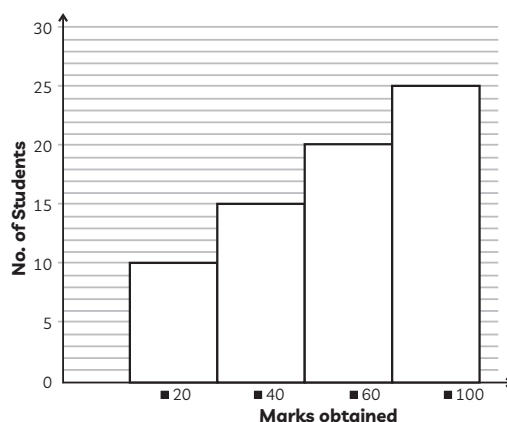
LONG ANSWER Type Questions (LA)

[4 & 5 marks]

7. The frequency distribution of the given data has been represented graphically as follows:

Marks obtained	Number of students
0 – 20	10
20 – 40	15
40 – 60	20
60 – 100	25





Do you think this representation is correct? Why?

Ans. No, here the widths of the rectangles are varying, so, we need to make certain modifications in the length of the rectangles so that areas are proportional to the frequencies.

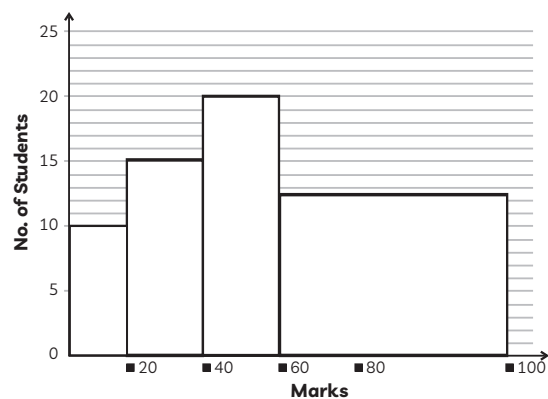
We proceed as follows:

- (1) Select a class interval with the minimum class size, here the minimum class size is 20.
- (2) The length of the rectangles are then modified to be proportionate to the class size 20.

Now, we get the following modified table

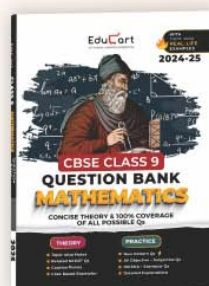
Marks obtained (Class Interval)	Number of students (frequency)	Width of the class	Length of the Rectangle
0 – 20	10	20	$\frac{20}{20} \times 10 = 10$
20 – 40	15	20	$\frac{20}{20} \times 15 = 15$
40 – 60	20	20	$\frac{20}{20} \times 20 = 20$
60 – 100	25	40	$\frac{20}{40} \times 25 = 12.5$

So, the correct histogram with varying width is given below



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